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ORIGINAL ARTICLE

Interpretation criteria for FDG PET/CT in multiple myeloma (IMPeTUs): final results. IMPeTUs (Italian myeloma criteria for PET USe)

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Abstract
FDG PET/CT (^{18}F-fluoro-deoxy-glucose positron emission tomography/computed tomography) is a useful tool to image multiple myeloma (MM). However, simple and reproducible reporting criteria are still lacking and there is the need for harmonization. Recently, a group of Italian nuclear medicine experts defined new visual descriptive criteria (Italian Myeloma criteria for Pet Use: IMPeTUs) to standardize FDG PET/CT evaluation in MM patients. The aim of this study was to assess IMPeTUs reproducibility on a large prospective cohort of MM patients.

Materials and methods Patients affected by symptomatic MM who had performed an FDG PET/CT at baseline (PET0), after induction (PET-AI), and the end of treatment (PET-EoT) were prospectively enrolled in a multicenter trial (EMN02)(NCT01910987; MMY3033). After anonymization, PET images were uploaded in the web platform WIDEN® and hence distributed to five expert nuclear medicine reviewers for a blinded independent central review according to the IMPeTUs criteria. Consensus among reviewers was measured by the percentage of agreement and the Krippendorff's alpha. Furthermore, on a patient-based analysis, the concordance among all the reviewers in terms of positivity or negativity of the FDG PET/CT scan was tested for different thresholds of positivity (Deauville score (DS 2, 3, 4, 5) for the main parameters (bone marrow, focal score, extra-medullary disease).

Results Eighty-six patients (211 FDG PET/CT scans) were included in this analysis. Median patient age was 58 years (range, 35–66 years), 45% were male, 15% of them were in stage ISS (International Staging System) III, and 42% had high-risk cytogenetics. The percentage agreement was superior to 75% for all the time points, reaching 100% of agreement in assessing the presence skull lesions after therapy. Comparable results were obtained when the agreement analysis was performed using the Krippendorff's alpha coefficient, either in every single time point of scanning (PET0, PET-AI or PET-EoT) or overall for all the scans together. DS proved highly reproducible with the highest reproducibility for score 4.

Conclusions IMPeTUs criteria proved highly reproducible and could therefore be considered as a base for harmonizing PET interpretation in multiple myeloma. A prospective clinical validation of IMPeTUs criteria is underway.

Keywords Multiple myeloma · FDG PET/CT · Interpretation criteria · Standardization · IMPeTUs

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Introduction

18F-FDG PET/CT (^{18}F-fluoro-deoxy-glucose positron emission tomography/computed tomography) is a functional imaging procedure with proven good performance in patients affected by multiple myeloma (MM). It may be considered as a valuable tool in the workup of patients with newly diagnosed and relapsed/refractory MM, in particular for the detection of paramedullary and extramedullary soft tissue masses or solid organ involvement. However, the major strength of FDG PET/CT is the ability to distinguish between metabolically active and inactive sites of the disease, a finding that makes this technique recommended by the International Myeloma Working Group (IMWG) as the actual "gold standard" method for evaluating and monitoring response to therapy [1]. Data derived from PET/CT images such as SUV max (standardized uptake value), number of hyper-metabolic foci, presence of extramedullary foci, and complete or incomplete FDG uptake suppression during or after therapy proved useful indexes to further stratify patients with different treatment outcome [2, 3].

However, interpretation issues in the evaluation of FDG PET/CT exist, which are particularly relevant in post-treatment PET scan images. For example, doubts may arise especially in the case of very recent bone fractures, vertebral collapses, recent metallic bone implants, low FDG uptake in lytic lesions, or increased and diffuse bone marrow uptake. There is, therefore, a need for standardization of PET readings to harmonize interpretation, especially in borderline cases. A number of heterogeneous interpretation criteria have been used in clinical trials by different MM cooperative groups based on semi-quantitative readout with variables SUVmax cut-off values, on visual assessment alone or on a combination of both methods, thus preventing data reproducibility [3–8].

Recently, different standard interpretation criteria have been proposed. Mesguich et al. [7] proposed some indications to interpret MM FDG-PET/CT in staging, during and after therapy, but these remain general suggestions. Other groups proposed SUV-derived parameters such as total lesion glycolysis (TLG) and metabolic tumor volume (MTV) [6] to assess the active disease burden at baseline and its variation (delta) as a consequence of therapy. However, a standard and widely accepted software program to harmonize MTV or TLG measure in clinical practice is still lacking. Furthermore, none of the proposed criteria have been clinically validated.

Recently, a group of Italian nuclear medicine experts, hematologists, and medical physicists defined new visual descriptive criteria (Italian Myeloma criteria for Pet Use: IMPeTUs) to standardize FDG PET/CT evaluation in MM patients [9]. These include the visual interpretation of images to quantify FDG uptake using the five-points scale of Deauville score (DS) proposed for interim and final FDG-PET in lymphoma [10], in association with a morphological and anatomical aspect of FDG distribution such as the bone marrow non-focal uptake, focal bone lesions (site, number and uptake), para-medullary, or extra-medullary lesions. These reporting criteria have been published in a recent and preliminary paper, in which IMPeTUs has been used to interpret baseline and end-of-treatment PET scan in a small cohort of 17 MM patients, with a good reproducibility among reviewers [9]. These criteria are only descriptive so far. Positivity cut-offs have not been set yet and will be recognized "a posteriori" basing on follow-up data in future works, with the same method employed for Deauville criteria in Hodgkin lymphoma [11, 12].

The aim of this study was to assess the feasibility in the routine clinical application of IMPeTUs criteria in a larger MM patient population and to define the reproducibility of the proposed reporting criteria by assessing the inter-observer agreement.

Materials and methods

The methodology used to draw the criteria has been described in a previous paper [9] where preliminary results regarding inter-observer agreement on 17 patients have been reported. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Briefly, a summary of methods follows:

Patients affected by symptomatic MM, enrolled in the multi-center, phase 3 EMN02 study [13] and scanned with FDG PET/CT at baseline (PET0), after induction (PET-AI) and the end of treatment (PET-EoT), and prior to the start of maintenance, were prospectively enrolled. All the patients signed an informed consent form.

Briefly, patients were randomized to receive four cycles of bortezomib-melphalan-prednisone (VMP) or bortezomib-cyclophosphamide-dexamethasone, followed by high-dose melphalan (HDM) and single or double autologous stem cell transplantation (ASCT) as intensification therapy. All patients received lenalidomide maintenance until progression or unacceptable toxicity. The PET images were a posteriori re-interpreted according to a blinded independent central review (BICR) methodology. Upon image upload, the web-based WIDEN® platform (DiXit, Torino, Italy) automatically distributed images to five expert reviewers (MB, AB, CN, MR, and AV, each one with more than 10 years of experience in oncological FDG PET/CT reading) who downloaded PET images into their own workstation and independently reviewed the scans according to the new IMPeTUs criteria by filling a standard on-line form for every patient. The report was considered complete when five out of five reviewers made their reviews.

All PET/CT scanners in which the PET images from patients enrolled in the EMN02 study were acquired underwent the qualification procedure for clinical trial adopted by FIL (Italian Lymphoma Foundation) in collaboration with AIMN (Italian Association of Nuclear Medicine), under the supervision of the FIL Core Lab at the Medical Physics Division of Santa Croce e Carle Hospital, Cuneo, Italy. Variability of SUV measurements in anthropomorphic phantom among PET/CT scanners was < 10%.

All PET/CT scans were acquired according to EANM PET procedure guidelines for FDG studies [14], but the following conditions had to be met: (a) patient scanning on full-ring PET/CT only; (b) iterative reconstruction applied to PET images with CT attenuation correction; (c) both anonymized CT and PET images available for central review; (d) PET scans of the same patient performed on the same scanner and (e) both PET and CT scans had to cover an enlarged whole body field of view including at least the region from the tip of the skull to the lower third of the femurs.

Additional quality control was applied to all scans centralized in the corelab. Here, scans were excluded from the analysis if: (a) the images were of poor quality with low statistics and not considered suitable for diagnostic interpretation, (b) the image data set was incomplete, and (c) large violation respect to uptake time and administered dose were found analyzing DICOM headers of PET scans.

All PET scans were reported using the IMPeTUs criteria. Briefly, the following parameters had to be checked: bone marrow metabolic state (BM), number and site of focal PET-positive lesions (Fx) with or without osteolytic characteristics (Lx), presence and site of extra-medullary (EM) or of paramedullary (PM) disease, presence of fractures (Fr). The degree of FDG uptake was visually quantified in the target lesion and extramedullary lesions according to the five-point Deauville scale adopted for interim and end-of-treatment PET scan in lymphoma. The criteria are reported in detail in Table 1.

Furthermore, semi-quantitative measures were also obtained in physiological areas corresponding to reference organs, liver, and blood mediastinal pool structures (BMPS) using a circular region of interest (ROI) with radius larger than 3 cm in the central portion of the liver far away from its edge, and a ROI within the aorta lumen but with a lower diameter of the vessel, taking care to avoid the vessel wall or areas of calcification for BMPS. The following semi-quantitative parameters were annotated and used to reinforce the visual analysis interpretation especially in borderline cases:

- Bone marrow SUVmax of the hottest lesion per macro-area
- Focal reference (REF) SUVmax of the hottest lesion per macro-area
- Focal reference SUVmean of the hottest lesion (5 pixels) per macro-area
- Fracture on REF: present/absent
- REF in EM: present/absent
- Liver SUVmax:
- Liver SUVmean:
- Mediastinal blood pool SUVmax:
- Mediastinal blood pool SUVmean:
- Comments to the exam:……

Statistics

The method used for inter-observer variability included the percentage of agreement and the Krippendorff's alpha [15]. This coefficient is 0 in the case of random coincidence and below 0 in the case of concordance lower than random coincidence (limits: −1; +1). Values for alpha statistics range from −1 to 1, and a rough guideline for interpreting the degree of agreement adapted from kappa statistics is as follows: < 0.00 = "total" disagreement, 0.00–0.20 = slight agreement, 0.21–0.40 = fair agreement, 0.41–0.60 = moderate agreement, 0.61–0.80 = substantial agreement, 0.81–1.00 = almost perfect agreement. The agreement was measured on each point of IMPeTUs criteria for each indication to the PET scan (PET0, PET-AI, and PET-EoT) and for all the indications together. Finally, the concordance among all the reviewers in terms of positivity or negativity of the FDG PET/CT scan was tested for different thresholds of positivity (DS 2, 3, 4, 5) for the main parameters (BM, FS, EM).

Results

One-hundred-three patients affected by MM were enrolled between February 2011 and April 2014 in the EMN02 study. Patient population characteristics are reported in Table 2. Briefly, median patient age was 58 years (range, 35–66 years), 45% were male, 15% of them were in stage ISS III and 42% had high-risk cytogenetics. The median follow-up was 24 months. The first 17 patients enrolled were excluded from the analysis because they belonged to the training set already published in a previous report [9], as preliminary test to explore the clinical applicability of the IMPeTUs criteria. Finally, 86 patients (237 FDG PET/CT scans, 84 PET0, 72 PET-AI, 81 PET-EoT) were included in this analysis.

Administered activity was 279 ± 68 MBq on average. The uptake time uniformly spanned the 58–88-min range. The average SUV in the liver was 2.3 ± 0.5. The difference for parameters among PET scans at different time points was never significant ($p > 0.60$).

Table 1 Summary of IMPeTUs criteria

Lesion type	Site	Number	Grading
Diffuse	Bone marrow		5-PS
	"A" if hypermetabolism in limbs and ribs		
F (Focal)	S (Skull)	X_1 (None)	5-PS
	SP (spine)	X_2 ($N = 1$ to 3)	
	Ex-Sp (extra-spine)	X_3 ($N = 4$ to 10)	
		X_4 ($N > 10$)	
L (Lytic)		X_1(None)	
		X_2 ($N = 1$ to 3)	
		X_3 ($N = 4$ to 10)	
		X_4 ($N > 10$)	
Fr (Fracture)	At least one		
PM (Para-medullary)	At least one		5-PS
EM (Extra-medullary)	At least one	N /EN (Nodal/Extranodal)*	

*For nodal disease: *C* cervical, *SC* supraclavicular, *M* mediastinal, *Ax* axillary, *Rp* retroperitoneal, *Mes* mesentery, *In* inguinal; For ENS: *Li* liver, *Mus* muscle, *Spl* spleen, *Sk* skin, *Oth* other)

5-PS Deauville 5-point scale

End of therapy and post-induction PET/CT PET/CT were carried out respectively 90 ± 10 days after ASCT and 15 ± 5 after induction.

The interobserver agreement was superior to 75% for all the criteria points, reaching 100% (Fig. 1) for skull lesions detection after therapy. Table 3 shows in detail the percentage of agreement in PET0, PET-AI, and PET-EoT for reporting bone marrow (BM) FDG uptake intensity, focal score (FS), extramedullary (EM) nodal (EM-N) or in extranodal (EM-EN) disease spread, number of focal lesions (F), number of lytic lesions (L), distribution of bone lesions (S: skull; Sp: spine; ExSp: extraspine), presence of fractures (Fr), involvement of bone marrow in limbs (A) and presence of paramedullary disease (PM). As a brief summary including all the indications, the concordance was ≥75% for BM, ≥ 76% for FS, ≥ 95% for EM, ≥ 76% for F, ≥ 77% for L, ≥92% for Fr.

Comparable results were obtained when the agreement analysis was performed through the Krippendorff's alpha method (due to the long list of concordance coefficients displayed in Table 4), either in every single time point of scanning (PET0, PET-AI, or PET-EoT) or overall for all the scans together (Table 5). As a brief summary including all the indications, the concordance was ≥0.23 for BM, ≥ 0.53 for FS, ≥ 0.06 for EM, ≥ 0.46 for F, ≥ 0.25 for L, ≥0.04 for Fr.

Finally, the concordance among all the reviewers in terms of positivity or negativity of the FDG PET/CT scan was tested for different thresholds of positivity along the Deauville five-point scale (score 2, 3, 4, 5) for the main parameters (BM, FS, EM). Concordance is reported in Table 6, showing that a score of 4 provides the highest agreement among all the reviewers.

Table 2 Patient population characteristics

No. of patients	86
Median age (range)	58 (35–66)
Median β2-micro (mg/l)	3.3
Median PLTs (× 10^9/l)	223
ISS stage III (% patients)	15
R-ISS stage III (% patients)	10
LDH > limit (% patients)	14.6
High-risk cytogenetics*(% patients)	42
Random HDM (% patients)	67
Random VMP (% patients)	33

*t(4;14) ± del(17p) ± del (1p) ± 1q gain detected by FISH

PLTs platelets, *ISS* International Staging System, *R-ISS* Revised International Staging System, *HDM* high-dose melphalan, *VMP* bortezomib, melphalan, prednisone

Discussion

In multiple myeloma, as well as in other neoplastic disorders, standardization in PET/CT interpretation is still an unmet need. In the present study, IMPeTUs criteria proved highly reproducible and suitable for routine reporting of FDG PET/CT in clinical practice. As a matter of fact, "very good" concordance was demonstrated among the five reviewers (expert in oncological PET/CT but only marginally trained for IMPeTUs criteria). All the heterogeneous morphological aspects of MM spread, such bone marrow FDG uptake, the number and site of hot focal lesions, the degree of lesion uptake, the presence of extra medullary and paramedullary disease, the presence of fractures and lysis at low-dose CT images were readily assessed by reviewers.

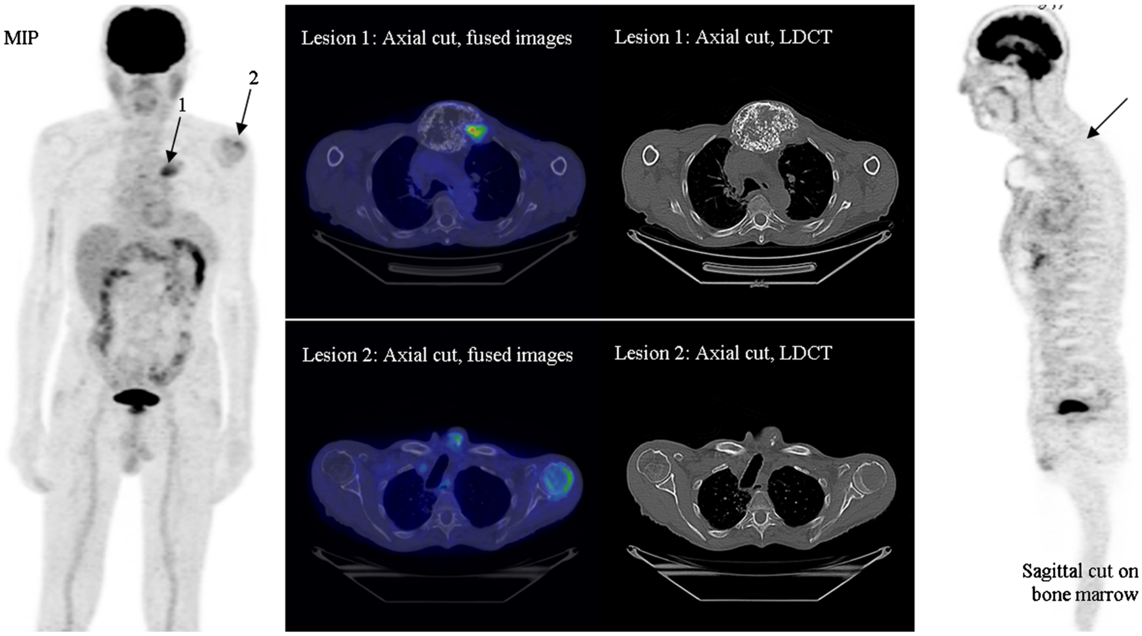

Fig. 1 EoT FDG PET/CT of a 77-year-old male patient and 100% concordance. All the reviewers agreed on the following description according to IMPeTUs: Bone marrow grade 2, no A parameter, F Ex-Sp 2 (2 extra-spinal hypermetabolic lesions), PM (paramedullary invasion of lesion 1), L2 (two lytic lesions). *Black arrows* on MIP indicate hot lesions. The *black arrow* on PET sagittal cut indicates the bone marrow uptake. *MIP* Maximum intensity projection; *LDCT* low-dose computed tomography

Similar to the Deauville Criteria implementation process, we proposed descriptive criteria based exclusively on visual interpretation. Despite that all the scanners belonging to the involved PET centers had undergone clinical trial qualifications, thus ensuring a variability of SUV measurements below 10%, we decided to avoid a semi-quantitative readout for PET scan interpretation (qPET). In fact, even a very low SUVmax variability, inferior to 10%, could generate interpretation disagreement in borderlines cases when a given semi-quantitative positivity cut-off is set. Furthermore, the majority of PET centers did not apply before study onset to any international qualification program such as EARL and the scanners adopted different image reconstruction algorithms. This would have resulted in highly variable SUV_{max} measurement, providing results that were not comparable and not reproducible [7, 16].

Visual interpretation avoids these sources of mistakes and is the optimal method for harmonizing PET interpretation across different PET centers. Nonetheless, in particular and difficult cases, SUV_{max} can be measured intra-patient to judge a lesion uptake in comparison to the mediastinum or liver, especially when the intensity of FDG uptake of the target lesion looks very similar at first glance to the reference organs. So, SUV max and mean can be considered optimal tools for reinforcing the reviewer visual interpretation if used in a relative way, in a single patient, to compare SUVmax of a given lesion to that of reference organ (MBPS or liver).

In this work, a slightly lower concordance among reviewers was found for the evaluation of bone marrow background uptake, since grade 3 (higher than mediastinum but ≤ liver) and grade 2 (≤ mediastinum) appear very similar. It is known that PET/CT has a low accuracy for the detection of bone marrow diffuse infiltration in MM patients due the high incidence of anemia and post therapy bone marrow rebound significantly reduce its specificity in this area [1].

A higher concordance among reviewers was found in the description of hypermetabolic bone marrow in limbs and ribs ("A" parameter on IMPeTuS). This is an uncommon finding

Table 3 Concordance among all reviewers for each single point of IMPeTUs criteria (value/100)

	BM	FS	EM	F	L	S	SP	ExSp	Fr	A	PM	EM-N	EM-EN
Staging (84 patients)	0.75	0.76	0.95	0.76	0.77	0.96	0.87	0.87	0.92	0.88	0.94	0.97	0.96
AI (72 patients)	0.76	0.76	0.97	0.76	0.77	0.99	0.86	0.89	0.94	0.90	0.94	0.98	0.99
EOT (81 patients)	0.76	0.84	0.97	0.84	0.86	1.00	0.89	0.93	0.98	0.84	0.99	0.99	0.98

BM bone marrow, *FS* focal score, *EM* extramedullary, *F* focal lesions, *L* lytic lesions, *S* skull lesions, *SP* spine lesions, *ExSp* extraspine, *Fr* fractures, *A* activated bone marrow in limbs, *PM* paramedullary, *EM-N* extramedullary –nodal localization, extramedullary-extranodal localizations

Table 4 Krippendorff's alpha: all reviewers for each single point of IMPeTUs criteria

	BM	FS	EM	F	L	S	Sp	ExSp	Fr	A	PM	EM-N	EM-EN
Staging (84 patients)	0.53 (0.41_0.66)	0.66 (0.56_0.76)	0.06 (−0.03_0.15)	0.56 (0.44_0.68)	0.54 (0.41_0.68)	0.08 (−0.06_0.21)	0.46 (0.33_0.60)	0.51 (0.36_0.66)	0.04 (−0.06_0.15)	0.45 (0.30_0.61)	0.58 (0.40_0.77)	0.23 (0.06_0.44)	0.04 (−0.06_0.15)
AI (72 patients)	0.33 (0.19_0.49)	0.53 (0.39_0.69)	0.25 (−0.02_0.54)	0.46 (0.30_0.63)	0.37 (0.20_0.55)	0.00 (−0.44_0.35)	0.42 (0.27_0.59)	0.60 (0.45_0.74)	0.33 (0.10_0.60)	0.21 (0.04_0.40)	0.46 (0.16_0.84)	0.22 (−0.42_0.71)	0.33 (−0.54_0.73)
EOT (81 patients)	0.23 (0.08_0.40)	0.58 (0.44_0.74)	0.41 (0.07_0.83)	0.56 (0.42_0.71)	0.25 (0.08_0.43)	0.25 (−0.70_0.70)	0.34 (0.17_0.52)	0.66 (0.53_0.81)	0.37 (−0.04_0.89)	0.19 (0.02_0.37)	0.66 (0.22_1.23)	0.51 (0.03_1.13)	0.32 (0.02_0.65)

BM bone marrow, *FS* focal score, *EM* extramedullary, *F* focal lesions, *L* lytic lesions, *S* skull lesions, *SP* spine lesions, *ExSp* extraspine, *Fr* fractures, *A* activated bone marrow in limbs, *PM* paramedullary, *EM-N* extramedullary –nodal localization, extramedullary-extranodal localizations

Table 5 Krippendorff's alpha all reviewers for all the indications together

	BM	FS	EM	F	L	S	Sp	ExSp	Fr	A	PM	EM-N	EM-EN
All (Staging + PostInd + EOT)	0.45 (0.36_0.53)	0.65 (0.58_0.72)	0.21 (0.06_0.38)	0.58 (0.51_0.66)	0.48 (0.40_0.57)	0.10(−0.01_0.21)	0.48 (0.39_0.56)	0.60 (0.53_0.68)	0.22 (0.08_0.36)	0.37 (0.26_0.50)	0.56 (0.43_0.71)	0.32 (0.12_0.56)	0.17 (0.04_0.30)

BM bone marrow, *FS* focal score, *EM* extramedullary, *F* focal lesions, *L* lytic lesions, *S* skull lesions, *SP* spine lesions, *ExSp* extraspine, *Fr* fractures, *A* activated bone marrow in limbs, *PM* paramedullary, *EM-N* extramedullary –nodal localization, extramedullary-extranodal localizations

Table 6 Krippendorff's alpha for all the reviewers with different cut-offs positivity. A threshold of 4 provides the highest agreement for all the indications

	Threshold 2			Threshold 3			Threshold 4			Threshold 5		
	BM	FS	EM	BM	FS	EM	BM	FS	EM	BM	FS	EM
Staging (84 patients)	−0.01	0.50	0.05	0.33	0.49	0.05	***0.46***	***0.54***	***0.09***	0.44		
AI (72 patients)	−0.01	0.47	0.25	0.21	0.47	0.25	***0.47***	***0.43***	***0.22***			
EOT (81 patients)	−0.02	0.52	0.40	0.19	0.59	0.40	***0.36***	***0.52***	***0.50***			

Threshold 5 is incomplete due to the low number of scans scored 5

BM bone marrow, *FS* focal score, *EM* extramedullary, *F* focal lesions, *L* lytic lesions, *S* skull lesions, *SP* spine lesions, *ExSp* extraspine, *Fr* fractures, *A* activated bone marrow in limbs, *PM* paramedullary, *EM-N* extramedullary –nodal localization, extramedullary-extranodal localizations

Bold Italic indicates the higher concordance

in reactive bone marrows, usually presenting with a homogeneous increase of FDG uptake concentrated in the spine and pelvic bones and thus could be considered a marker of MM harbinger in bone marrow. However, this should be clinically validated. A difference between score 2 or 3 is therefore likely to be clinically irrelevant.

It is important to point out that a good agreement with such a method is really complicated to reach in the absence of positivity cut-off. The proposed image reading is played, in fact, on a five-point scale for each parameter and not just on a binary output (positivity-negativity). Once the positivity cut-off will be set, a binary output will be possible and the concordance will tend to be perfect. So, despite a non-perfect agreement in some cases, the concordance reached is, in our opinion, impressive.

The highest agreement of end-of-therapy exams is probably related to the fact that, being the therapy really effective, a consistent number of patients have a negative exam.

Another relevant aspect to discuss is the use of Krippendorff's alpha coefficient. Although more difficult to interpret in comparison to a standard percentage of agreement, this method is more indicated because automatically corrects for a casual agreement between reviewers, which is higher if the number of observations is low.

Once again, we want to stress that that IMPeTUs criteria are not yet ready for clinical application. We need, in fact, to set positivity cut offs for each single point described (BM, F, EM, and so on) and this will be possible upon the median follow-up time for the entire cohort of patients included in the present analysis will reach a median of 2 years.

So far, we have tried to set a positivity cut-off based only on reviewers' concordance. We have found that score 4 (uptake higher than liver) for bone marrow, focal score, and extra-medullary disease leads to the higher concordance for all the indications to PET (staging, post induction and end of therapy). This does not have a clinical meaning yet and must be considered as a very preliminary result in the field of pure statistical observation. Interestingly, this is in accordance with positivity criteria proposed but not validated by other groups [7, 17].

Furthermore, the higher concordance among reviewers for score 4 at the end of therapy scan is a relevant result. Therapy assessment through FDG PET/CT, in fact, has a strong impact on the subsequent disease management not only because it provides a prognostic index and a further patient stratification but also because it can drive therapeutical choice and administration.

One drawback of IMPeTUs is the number of analyzed parameters. Including a high number of parameters was mandatory to start the analysis on such a complex disease. It is very likely that, upon reaching an adequate time of follow-up, these criteria could be simplified by deleting non-prognostic features to come to a simple, standardized, and validated PET reporting system.

Conclusions

In this work we have proven that IMPeTUs descriptive criteria are highly reproducible and can be considered as a strong base for harmonizing PET interpretation in multiple myeloma patients. Reviewers had the greatest agreement for parameters scored 4 according to DS. Future studies to compare IMPeTUs with patients follow-up are needed to assess positivity cut-offs and exclude non-prognostic parameters.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflicts of interest.

Research involving human participants All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent All the patients signed a specific informed consent.

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